

WASP-50b

R-1692

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131213 · created 2026-08-02T13:17:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456639.807 +/- 0.024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.028
Transit depth (Rp/Rs)^2	1.07 +/- 0.58 [%]
Orbital Inclination (inc)	89.69 +/- 2.36
Airmass coefficient 1 (a1)	1.17 +/- 0.014
Airmass coefficient 2 (a2)	-0.0999 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	None
Optimal Aperture	3.48
Optimal Annulus	9.04
Transit Duration (day)	0.083 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.028 vs 0.1404 (deviation 1.34σ)
Duration fit vs published	0.083 d vs 0.0754 d (deviation 0.76σ)
Mid-time O–C	40.3 min
Depth SNR	3.7
Residual scatter	1.16 %

Light curve

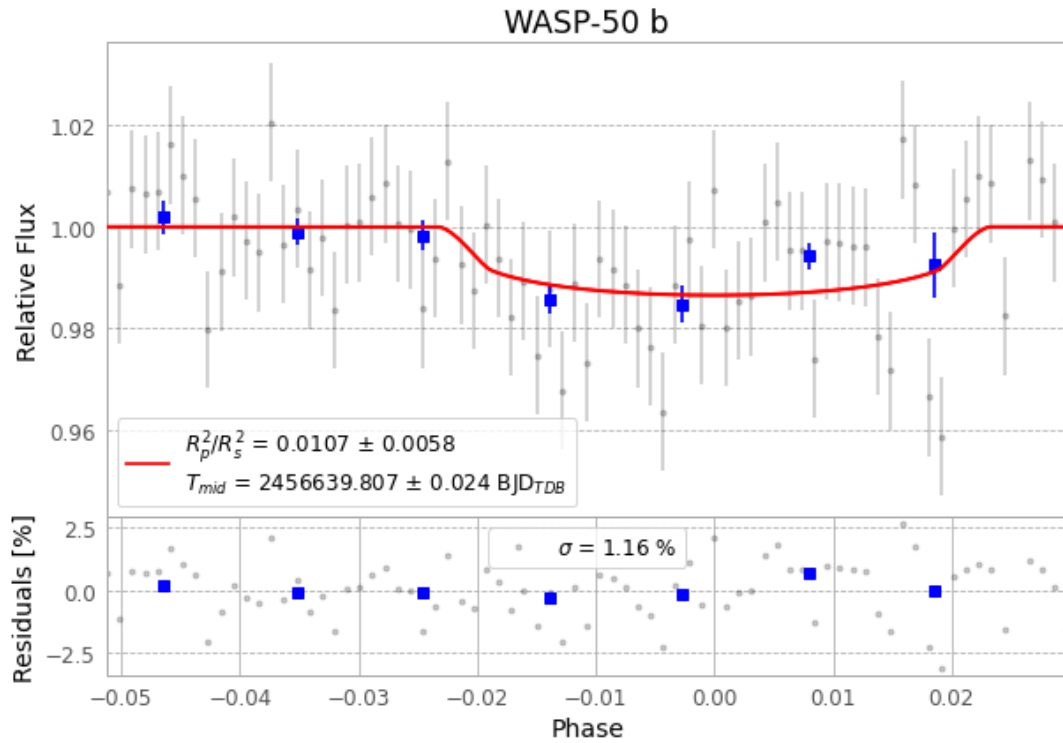


Figure 1 — FinalLightCurve_WASP-50 b_2013-12-12.png (SHA-256 a019a8b3999da5f4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 256], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4cce0509839a259e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50131213045112.FITS → WASP-50131213083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131213.log (6635ee20ef40...), FinalLightCurve_WASP-50 b_2013-12-12.png (a019a8b3999d...), FinalLightCurve_WASP-50 b_2013-12-12.csv (7f3e5cdce1f5...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 13:17Z.